# SEÁN Ó MURCHÚ

## Senior Software Engineer

sean@example.com +353 1 555 0100 Tallaght, Dublin, IE seanomurchu.dev
seanomurchu seanomurchu seanomurchu @seanomurchu @sean@fosstodon.org
@seanomurchu seanomurchu @seanomurchu talk recording

Backend engineer with eight years of experience designing distributed, event-driven systems. Specialises in functional programming, observability, and developer experience.

## EXPERIENCE

### Senior Software Engineer

Acme Corp

Jun 2022 – Present Dublin, Ireland

*Platform team lead. Owns the event-sourcing stack underpinning the wider product surface.*

- Led the migration of a customer-facing monolith into a set of event-driven microservices, halving p99 latency.
- Designed and rolled out an event-sourcing platform now used by four product teams.
- Mentored three engineers from mid to senior level.

### Software Engineer

Liffey Labs

Jun 2019 – Jun 2021 Remote

- Built and shipped the first version of a SaaS product from scratch alongside a two-person team.
- Introduced the CI/CD pipeline and developer-onboarding flow that scaled the engineering org from 3 to 15.

## VOLUNTEER

### Mentor

CoderDojo Dublin

Jun 2020 – Present

*Volunteer-led coding club introducing young people to programming.*

- Weekly mentoring sessions for 10–14 year-olds learning to code.
- Curate the Scala / functional-programming track.

## PROJECTS

### Application

### Hyperion

Acme Corp

Lead Architect

Jan 2024 – Aug 2024

*Distributed task scheduler in Rust*

- Handled 10k tasks/s on a single node.
- Designed an idempotent retry protocol.

Rust Tokio gRPC

## AREAS OF FOCUS

- Distributed systems: Kafka, CDC, idempotent consumers, event sourcing.
- Functional programming and type-driven design in Scala and Haskell.
- Developer experience: tooling, CI/CD, observability.
- Engineering culture: ADRs, RFCs, mentoring.

## SKILLS

**Languages** – Scala Haskell Python Go Rust

**Infra** – Kafka AWS Terraform Docker Kubernetes

## LANGUAGES

English ●●●●●●

Irish ●●●◐○○

French ●●●●○○

Portuguese ●●◐○○○

## EDUCATION

### M.Sc. in Computer Science

Tallaght Institute of Technology

2017 – 2019

Distinction

Distributed Systems Type Theory Concurrency

### Computer Science

Trinity College Dublin

2014 – 2017

## CERTIFICATIONS

### Certified Kubernetes Administrator

Aug 2023 CNCF

### Crucible

Maintainer

Sep 2024 – Present

*Migration tool for legacy databases*

- Schema diffing across PostgreSQL and MySQL.

### Library

### Tinkerbell

*Tiny IRC bot*

Lua

### Other

### Quill

## PUBLICATIONS

### Articles

- *Event Sourcing in Practice*.
  Personal Blog · 14 Jun 2024
  *A walk-through of idempotent Kafka consumers and the replay strategy that ships with them.*
- *Untyped Note*.
  Jan 2024

### Conference Papers

- *Idempotent Kafka Consumers at Scale*.
  EuroSys · Apr 2024

### Talks

- *Designing for Failure*.
  ScalaIO · Sep 2023

### Books

- *Functional Domain Modelling*.
  Self-published · 2022

## INTERESTS

Music – Trad Jazz

Sport – Hurling Climbing

Travel – Japan Iceland

### Certified Kubernetes Application Developer

Jan 2024 CNCF

### AWS Solutions Architect — Professional

May 2023 AWS

### AWS DevOps Engineer — Professional

Sep 2024 AWS

### Hashicorp Terraform Associate

Dec 2022 Hashicorp

### Certified Kubernetes Security Specialist

Mar 2024 Linux Foundation

### Google Cloud Professional Architect

Nov 2023 Google Cloud

## AWARDS

### Best Paper Award

ScalaConf

Sep 2024

*For* Idempotent Kafka Consumers.

### Dean's List

## REFERENCES

### Dr Ada Lovelace

*Seán is a calm, methodical engineer who left every system he touched in a better state than he found it. Recommended without reservation.*

*Pragmatic, curious, unfailingly kind — would hire again in a heartbeat.*